

# Report B — Automation Report

*What we automated, what it can do, what is new, and how to grow it.*

Automation means: a student or mentor does not have to click Load → Preprocess → Reconstruct for every file. One command (or one GUI button) finds files, picks scans, runs the same scientific pipeline, scores the result with written thresholds, and writes a report. The science is the same as the GUI. The novelty is the unattended path around it.

*Not a clinical diagnosis. Pass/fail is a research quality gate, not a tumor diagnosis.*

## 1. What problem automation solves

UM-BMID-style cubes can contain hundreds of scans. Doing each one by hand is slow and uneven (different people pick different beamformers). Automation makes the experiment repeatable: same geometry grid, same ROI rules, same score weights, same report layout.

- Discover measurement vs metadata files under one or more folders.
- Pair `md_list_*.mat` with `fd_data_*.mat` so people do not upload labels without data.
- Choose a diverse subset of BMID scans (healthy + small/medium/large + location/BIRADS/shape).
- Run load → preprocess → DAS/DMAS/DMAS-D4 → ROI spots → tumor-candidate → characterization → confidence.
- Classify each case pass / warning / fail / blocked using a named profile.
- Write `latest_auto_analysis.md`, JSON evidence, and showcase PNG figures.
- Optionally open the best tumor-candidate case in the GUI.

## 2. How you can run it (three doors, one engine)

All three call `AutoValidationService`. That is important: CLI, GUI, and MCP cannot drift apart.

- **CLI:** `python tools/run_auto_validation.py --profile balanced`
- **GUI:** Acquisition → Run Auto Validation (scope: 8 diverse scans by default).
- **MCP:** `automation/mcp_server.py` exposes the same job over stdio for agent tools.
- Related helper: `python tools/auto_intake_agent.py` checks files before a person uploads them.

Default roots prefer `microwave/umbmid/simple-clean` when that folder exists. Otherwise `project/datasets` is used (that is why a recent run validated `sample_touchstone.s2p`).

## 3. Capability map — module by module

### 3.1 Discovery (`automation/discovery.py`)

Walks folders. Classifies each file as measurement, metadata, or unknown. `fd_data_*.mat` is a UM-BMID cube. `md_list_*` / `metadata_*.mat` is labels only. `.s2p/.s1p/.s4p/.s8p` and other `.mat` files are measurements. Unknown extensions are skipped. Ranking prefers `fd_data_` files first so real cubes beat leftover samples when both exist.

### 3.2 Resolver (`automation/resolver.py`)

Turns files into targets: (measurement path, optional scan index, optional metadata path). Metadata-only files are useful only if a paired `fd_data` file can be found beside them or in the repo. BMID strategies:

- **diverse** (default) — about 8 scans: one healthy, one small, one medium, one large, then extra quadrants / BIRADS / shapes.
- **all** — every scan up to `max_bmids_scans` (0 means unlimited; this is slow for ~2000 scans).

- **tumor\_and\_healthy** — strongest labeled tumor plus one healthy.
- **first / first\_healthy** — simple debugging picks.

Tumor ranking when not using diverse: larger diameter first, then an offset preference near ~3 cm so huge edge phantoms are not always chosen.

### 3.3 Per-target pipeline (automation/pipeline.py)

This is the automated twin of the GUI workflow.

- Load with `load_dataset(path, scan_index)`.
- Read ground-truth tumor (x,y) from metadata if present.
- Preprocess: BMID → all methods none, Week 3 off. Otherwise → Savitzky–Golay window 7 poly 3, Week 3 on, clutter on, global normalize on.
- Reconstruct DAS, DMAS, DMAS-D4 on a 64×64 grid. BMID FOV forced to ±6 cm; phase-delay radius off for comparable batch runs.
- Select beamformer: if GT exists, pick the image whose ROI is closest to the labeled tumor; else weighted quality.
- Detect up to 4 ROIs (threshold 0.7 internally, `min_score_ratio` 0.28 in automation). If GT exists, keep the nearest hotspot even if NMS dropped it.
- High-resolution ROI reconstruct (min 96<sup>2</sup>, up to 256<sup>2</sup>, ×4 upscale of the coarse box).
- Characterize size/SCR/FWHM; assess confidence; write figures.
- Classify with the chosen threshold profile.

### 3.4 Quality profiles (automation/policy.py + classifier.py)

Hard fails: load error, non-finite image, energy below min, ROI area below min. Soft warnings: localization error, SCR, SNR, confidence, or “GT tumor but not a candidate”. Blocked: cannot even start (no paired measurement).

| Gate                    | strict | balanced | lenient |
|-------------------------|--------|----------|---------|
| min energy              | 1e-10  | 1e-12    | 1e-14   |
| min ROI area (px)       | 8      | 4        | 1       |
| max loc. error (cm)     | 1.5    | 3.0      | 5.0     |
| min SCR                 | 5.0    | 2.0      | 0.5     |
| min SNR                 | 1.0    | 0.5      | 0.1     |
| min confidence          | 0.55   | 0.35     | 0.20    |
| require candidate if GT | yes    | yes      | no      |
| finite output required  | yes    | yes      | yes     |

Warning cases can still be recommended\_upload if localization is within 1.25× the error gate. That lets slightly blurry but useful scans into a demo set.

### 3.5 Reports, cache, figures, MCP

- `reports.py` writes `results/latest_auto_analysis.md` (tumor index, detection table, showcase, walkthrough) and `latest_auto_validation.json`.
- Showcase pick: labeled tumor first, then strongest candidate, then smaller localization error, then higher confidence.
- `figures.py` copies selected / beamformer-compare / ROI-refine PNGs.
- `cache.py` avoids re-running identical (file, scan, profile) work in one session.
- MCP server is a thin stdio wrapper so an external agent can start the same job.

## 4. What automation is capable of today (plain list)

- Batch-screen a folder of .mat / .s2p files without opening each GUI tab.
- Tell you which BMID indices are labeled tumors before reconstruction (tumor index).
- Produce Yes/No tumor-candidate plus a reason sentence per case.

- Report localization error in cm against laboratory GT when labels exist.
- Compare DAS vs DMAS vs DMAS-D4 automatically.
- Keep a JSON audit trail of metrics, taxonomy, preprocessing steps, and ROI spots.
- Fail closed on corrupted files instead of crashing the desktop app.
- Run load-only mode (--no-reconstruction) for a fast finite/energy sanity check.

What it does **not** do: it is not a trained neural classifier; taxonomy is metadata buckets. It does not claim BIRADS medically. It does not replace a radiologist. It does not search every geometry on every scan (that is Auto Tweak in the GUI; batch uses a fixed conservative geometry).

## 5. Novelty this automation brings to the project

Many microwave-imaging student projects stop at “we plotted one DAS image.” The new layer is an experiment operating system around the physics:

- **Format-aware batching:** the same job understands Touchstone S-matrices and BMID cubes, including “do not preprocess \_adi”.
- **Label-aware scan picking:** diverse strategy is not random; it forces healthy + size + location coverage so a demo is honest.
- **Physics-in-the-loop scoring:** gates are localization (cm), SCR, SNR, ROI area, energy — not a black-box accuracy percentage.
- **Multi-spot ROI:** more than one hotspot can be reported; GT is used to keep a nearby spot, not to hide other spots.
- **One evidence pack:** markdown for humans, JSON for machines, PNG for slides, GUI handoff for inspection.
- **Profiled strictness:** mentors can ask for lenient triage or strict localization without changing code.
- **Three interfaces, one pipeline:** students, GUI users, and MCP agents cannot silently use different math.

## 6. How this sits next to the rest of the software

Manual GUI path remains the teaching path: you see every tab. Automation is the verification path: you prove the same modules work on many files. Auto Tweak (reconstruction/auto\_calibrate.py) is a different automation: it searches antenna angle, flip, 355° vs 360°, phase-delay radius, and ROI mode for **one** loaded scan. Batch validation does not currently call Auto Tweak, on purpose, so a 8-scan run finishes in a reasonable time and stays comparable.

## 7. How we can develop it further

### Near term (practical)

- Default Module 3 input to Hamming / Week-3 time-domain when those arrays exist (README already lists this gap).
- Optional “Auto Tweak lite” inside batch: only on showcase tumor cases, not on every healthy scan.
- Write a CSV summary (one row per scan) for Excel mentors who will not open JSON.
- Store per-scan reconstruction settings in the JSON so a GUI reload is bit-identical.
- Add a dry-run that only prints the tumor index and chosen diverse scans (no beamforming).

### Medium term (science)

- Replace the placeholder severity mapping with a trained model that uses image features + metadata, still labeled research-only.
- Learn suspicion\_threshold per dataset family instead of a global 0.45.
- Support empty-chamber (\_emp) subtraction when both adi and emp cubes are present.
- Quantify false-alarm rate on healthy scans as a first-class metric in the analysis report.
- Add a third data family (e.g. time-domain .mat already IFFT’d) with an explicit skip-IFFT flag.

### Longer term (product / teaching)

- A simple “lab exam mode”: hidden GT, student reconstructs, automation grades localization only.
- Cloud/batch runner that processes the full 2264-scan inventory overnight and publishes a dashboard.
- Uncertainty maps (pixel-wise coherence factor) exported beside the image.
- Keep the medical disclaimer in every new report surface so nobody treats scores as diagnosis.

## 8. Beginner cheat-sheet

- If you have one .s2p: automation will load it, clean S21, reconstruct, and likely say tumor-candidate Unknown (no GT).
- If you have UM-BMID: put fd\_data and md\_list together; run diverse profile balanced.
- Read section 1 of latest\_auto\_analysis.md to find tumor scan indices.
- Read section 2 for Yes/No and localization error.
- Open the showcase image instead of clicking every tab.
- Use lenient only when you are debugging failures; use strict before a demo you care about.